

Final Project—MTH 602

Goal All students in this class have scientific interests, whether formal (e.g., thesis or dissertation projects) or more informal (personal or hobby-related). The aim of this project is to explore how scientific machine learning (SciML) can help address traditional scientific problems. You will select an existing scientific question or area of interest and investigate how machine learning can be applied in that context.

The intent is twofold: (i) to give you an opportunity to study something directly relevant to your own research or interests, and (ii) to help your classmates learn about new scientific areas and machine learning techniques through your project.

Overview Students should form groups of 1–3 members (2 is preferred). Each group will design and carry out a project of their choice. An acceptable project must meet the following criteria:

1. It is a well-motivated and clearly defined scientific endeavor.
2. It applies machine learning in a meaningful way.
3. Projects should not just apply a “black box” model but should connect SciML methods to the scientific structure of the problem.
4. You can consider a “survey-style” project (e.g., a comparative study of ML approaches for a given problem).

You will propose both the topic and the goals you hope to achieve. The project should be focused and manageable, while still aiming to be ambitious. A particularly strong outcome would be if your project contributes concretely to your dissertation or thesis research. You will be expected to work on the project throughout the semester, with regular check-ins.

Deadlines

- **Wednesday, September 30:** Submit (via email) a brief project description (a few sentences is sufficient) covering (i) the scientific question you are addressing, (ii) limitations of classical methods that machine learning may help overcome, and (iii) the SciML techniques you expect to explore. Your project must be approved before you begin work.
- **Week of October 12:** Short presentations on the science behind your project and how you plan to use SciML. Please also submit (or include in your talk) a project plan/timeline (with milestones). Presentations will either be in-class or pre-recorded.
- **Monday, November 30:** First draft of project report due (for feedback, not graded). Your report will get feedback from the instructor and other students in the class.
- **Wednesday, December 9:** Final presentations and written reports due.

Grading and Guidelines Your grade will be based on both presentations and the written report.

Presentations (50%). Each presentation should:

1. Clearly define the project and its goals.
2. Explain why the topic is interesting (e.g., applications or significance).
3. Summarize any relevant mathematical or algorithmic details.
4. Mention any publicly available implementations of related algorithms/code.
5. Outline your work plan and progress.
6. Code demos can be used as necessary.

Requirements:

- Each group presents for ~ 10 minutes.
- All group members must speak for roughly equal amounts of time.
- Presentations should be rehearsed and fit within (but not exceed) the time limit.
- Use of slides (Keynote, Beamer, PowerPoint, etc.) is strongly recommended.

Presentations that meet these requirements and demonstrate clear effort will receive full credit.

Report (50%). The final report should be written as a scientific paper. The suggested structure is:

1. (~ 1 page) Introduction: overview of the project and its goals.
2. (1–2 pages) Background: why the science is interesting and the current state-of-the-art.
3. (2–5 pages) Methodology: technical background and approach.
4. (~ 5 pages) Results: tests, validation, data, scaling, and key findings.
5. (1–2 paragraphs) Conclusion: summary, observations, and possible future directions.

Reports may be written in Jupyter notebooks or in $\text{\LaTeX}$. They should include:

1. Figures and plots.
2. Mathematical expressions.
3. Citations to prior work.

Reports should not:

1. Exceed 20 pages (figures and references included) unless absolutely necessary. **Target length: ~ 10 pages.**
2. Contain source code (except for illustrative snippets). Code should be maintained in a git repository.

The report should be written in clear scientific prose, as if prepared for journal submission.

Code.

Your projects should be tracked by git (using either github, gitlab, or bitbucket). Students should strive to make their code and data reproducible (within reason). Projects should include a README.md and instructions for running the code.

0.1 Previous projects

Some examples include...

- Denoising functional data using variational autoencoding neural operator
- Kolmogorov-Arnold Networks for surrogates
- Parameter estimation using normalizing flow for overlapping GW signals
- Project Lacuna, a framework for quantifying and classifying missing data mechanisms in scientific datasets
- Implementation of ML-accelerated conservation scheme in FLASH: hybrid numerical + machine learning approach
- Using autoencoders to speed up the existing surrogate modeling algorithm and extract non-linear combinations of parameters that better represent the data
- Learning simulation error